

Mohammed Mokhtar Soliman Salama

+20 115 036 8489 | muhammedmukhtar822@gmail.com | [linkedin.com/in/mohammedmokhtar2](https://www.linkedin.com/in/mohammedmokhtar2)

github.com/Mukhtar82 | Portfolio

EDUCATION

Helwan National University

B.Sc. in Electrical Engineering – Intelligent Systems Major

oct. 2022
Cairo, Egypt

- **Coursework:** Data Structures, Algorithms, Databases, Computer Systems, Machine Learning
- **Activities:** Student Union Chair – Family Committee , Head of Problem-Solving committee – Pixels Egypt

SKILLS

Languages : Python, C/C++, JavaScript (React.js, Next.js), HTML/CSS, SQL (PostgreSQL, MySQL)

AI Frameworks/Tools : TensorFlow, Keras, PyTorch, Scikit-learn, LangChain, Hugging Face

Tools and Platforms : Git, Jupyter, OpenCV, Docker, Fast API, LLMs, DL, ML, NLP, CV, RAG

EXPERIENCE

IT Engineer

Helwan Tech Plus (Helwan National University)

July 2025 – Feb 2026
Cairo, Egypt

- **Engineered and deployed a RAG-based AI assistant** using **Llama 3** and **FastAPI**, creating a centralized knowledge hub that allows students to instantly retrieve academic files and university policies.
- **Reduced administrative workload by 60%** by automating responses to routine inquiries, significantly cutting down call center volume and queuing time for university staff.
- **Optimized the student experience** by providing 24/7 self-service access to critical information, eliminating the need for physical visits or phone calls.

AI Engineer Intern Link

Information Technology Institute (ITI)

Jul. 2025 – Sept. 2025
Cairo, Egypt

- **Implemented Evolutionary Computing strategies**, specifically **Genetic Algorithms**, to solve complex non-linear optimization problems where traditional gradient-based methods are insufficient.

AI Researcher Intern

Qayedny

Jul. 2025 – Aug. 2025
Cairo, Egypt

- **Orchestrated an automated recruitment workflow** using **LangGraph**, **n8n**, and **Airtable**, integrating a custom CV parser that extracts candidate skills and auto-fills application forms to eliminate manual data entry.
- **Developed an intelligent matching system** that connects students with startups by analyzing parsed resumes against requirements, streamlining the hiring process for both parties

PROJECTS

NutqAI (Graduation Project)

- Designing and implementing an intelligent **Arabic** speech therapy system using AI and NLP techniques to support children with speech impairments, featuring real-time voice analysis, pronunciation feedback, and adaptive learning modules.

Official University Chatbot

- Engineered a production-ready RAG chatbot for the official university website using **Gemini API** and **ChromaDB**, providing students with instant, automated access to academic resources.
- Deployed the solution using **FastAPI**, **Docker**, and **Hugging Face Spaces**, ensuring scalable, 24/7 availability and low-latency responses for concurrent users.

Risk AI: Industrial IoT Safety System Link

- Engineered a real-time hazard detection system using **ESP32** and predictive algorithms to forecast gas leaks with 2-second latency.
- Developed a **React** dashboard visualizing live sensor streams (CO2, Temp) via high-performance Recharts.

NeuroCaption: Multimodal Image Captioning [Link](#)

- Designed a unified AI pipeline integrating **ResNet** (CV) and **GPT-2** (NLP) to generate semantic descriptions for input images.
- Deployed a decoupled architecture with a **FastAPI** inference engine (Dockerized) and a **Next.js** frontend on Vercel.

VitalSense: AI Health Prediction Platform [Link](#)

- Built a full-stack diagnostic system utilizing Machine Learning models to analyze vital signs (Heart Rate, BP, Oxygen).
- Orchestrated a **REST API** architecture connecting a Python backend to a responsive React interface for real-time predictions.

Certifications

HCIP AI Training [Link](#)

Aug. 2025 – Nov. 2025

Huawei Academy (portsaid university)

Cairo, Egypt

- **Applied AI Domains:** Acquired specialized skills in **Computer Vision** (Image Preprocessing, Feature Extraction), **Speech Recognition** (GMM-HMM, DNN), and **NLP** (Text Vectorization, Language Models).

NLP Engineering Training [Link](#)

Jul. 2025 – Aug. 2025

National Telecommunication Institute (NTI)

- **Optimized language models** by applying advanced preprocessing techniques (TF-IDF, N-grams) and **fine-tuning** pre-trained models to improve context understanding and similarity detection.

Computer Vision Engineering Training [Link](#)

April. 2025 – Jul. 2025

National Telecommunication Institute (NTI)

- **Implemented high-performance vision pipelines** for object detection and face recognition using **PyTorch**, **TensorFlow**, and **OpenCV**.
- **Enhanced model accuracy** by leveraging **Transfer Learning** on state-of-the-art architectures like **ResNet** and **EfficientNet** for complex classification and segmentation tasks.

Data Engineering Training [Link](#)

April 2024 – Oct 2024

Digital Egypt Pioneers Initiative (DEPI)

- **Architected scalable data pipelines** using **SSIS** and **Azure**, ensuring data integrity and availability for warehousing and analytics.
- **Implemented MLOps best practices** using **MLflow**, bridging the gap between model development and production deployment.

IBM AI Engineering Professional Certificate [Link](#)

Feb. 2026

Coursera

- It consists of 11 Courses and 2 Projects.

More Certificates [Link](#)

Volunteering

Head of Problem-Solving Committee

Sept. 2023 – Aug. 2025

Pixels Egypt org.

Cairo, Egypt

- Trained volunteers in algorithms and data structures; Communication and Presentation Skills
- Trained 250+ University Students all around Egypt in algorithms and data structures; organized contests and events.

Deputy Organizer

Sept. 2023 – April 2026

Resala Charity Organization

Cairo, Egypt

- Led medical camp logistics to serve thousands in underserved Egyptian villages.

Student Union Chair Family Committee

Sept. 2024 – Present

Helwan National University

Cairo, Egypt